



**ed2k Protocol Documentation
Client - Serveur
Version 3.0**

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1- General Purpose

1.1- Server's functions

The eDonkey server doesn't really do anything. En tout cas it has not lots of functionalities, it just have to support a great number of connections. Its main task is to keep and manage clients connections information ; such as :

- How to join us ?
- Their shared files listing
- Potentially become a communication tunnel between two clients in case of a LowID <-> HighID communication (see more in section 1.3)

Next it has to answer to a certain number of requests such as "login requests", and especially file searching ones. We will see later that in this case, the server has to return a clients list for who has the requested file.

That's all.

1.2- Communication Ports

4661 : Main communication port : client <-> serveur

All requests pass through this one.

4662 : Communication port between two clients or one client and the server for ID assignment.

4665 : Communication port for a LowID <-> HighID connection.

For more about ports usage, go to the next section.

1.3- LowID and HighID

How does the server assigns the ID ?

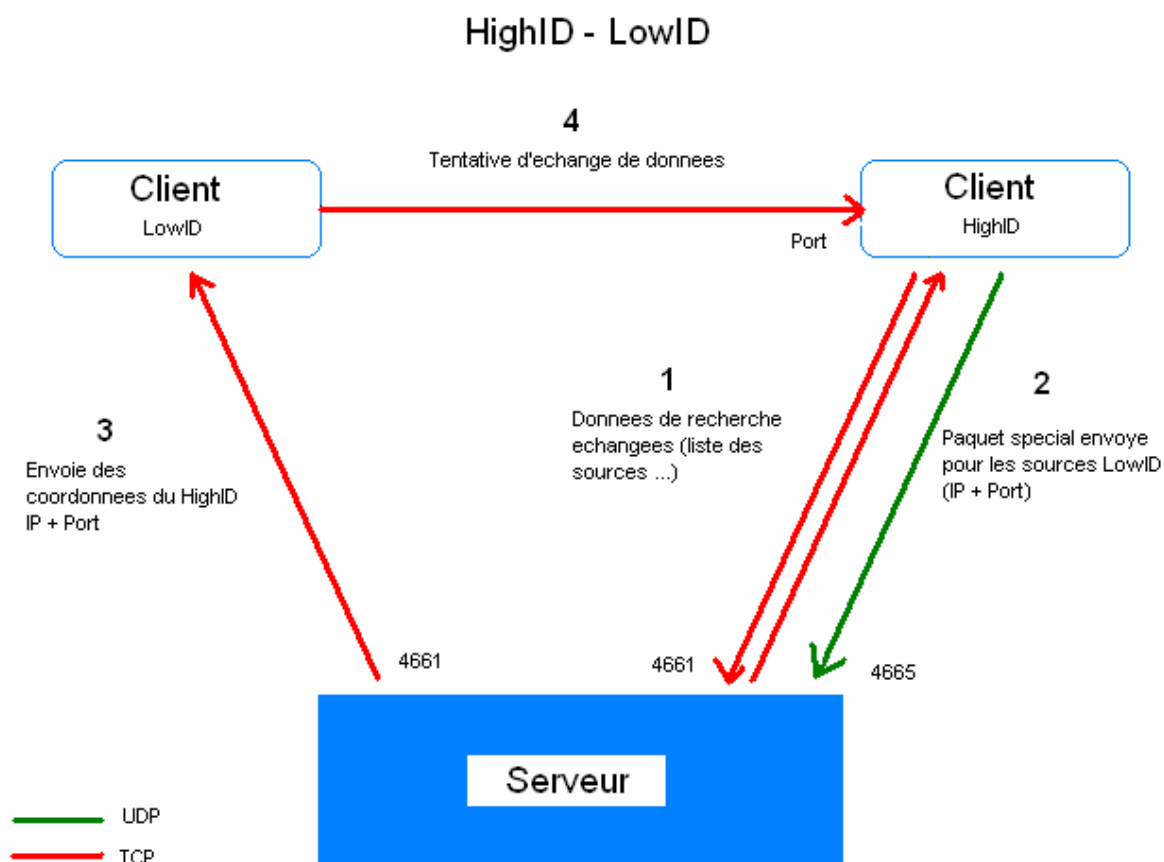
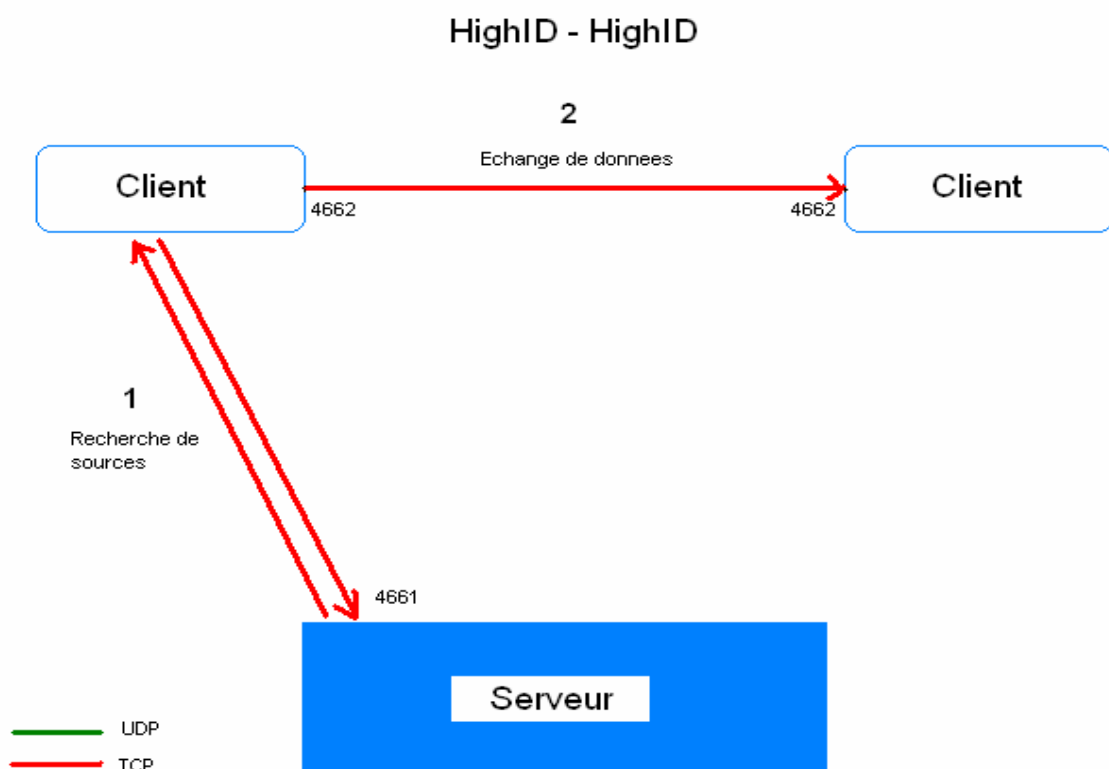
It's very easy, when a connection has been accepted, the server receives the listening port's number for the client, which default value is 4662.

Once this information has been collected, the server will first check if the client is not already connected by doing comparisons between the new client's IP/port and the IP/port of clients which are connected yet. If you are already connected, the server will automatically assign you a Low ID ; else the server will try to establish a TCP/IP connection on the client's listening port.

So there are two possibilities :

- The client is reachable : the server assigns a HighID
- The client is not reachable for any reason : the server assigns a LowID

How does the different types of communication evolve ?



How does the ID is calculated ?

For a HighID :

We'll take the following IP as an example : 81.56.25.36

Convert it in hexadecimal : 0x51.0x38.0x19.0x24

Invert the datas byte by byte : 0x24.0x19.0x38.0x51

Concatenate all : 24193851

And translate it as an integer : 605632593

You just got the HighID

For the reason that this operation is reversible, the server is able to determine an IP address just from a HighID.

For a lowID :

The server creates itself a factice IP such as : NumUser.0.0.0 where NumUser represents the number of clients in the server's list, for example 100.0.0.0 when there are 100 users. Then it does the same as if it was generating a HighID on this factice IP.

Why could you get a LowID :

In most cases, you've a Firewall which is not correctly configured. So you have to change its configuration to permit communications threw the port which receives the data from the machine you are connected to.

In some other cases, on the lugdunum's servers, when you disconnect and connect again but too fast, the server will assign you a LowID even if you had a High one before the disconnecting.

That's due to a latency in the sever's management of the users list.

So you must disconnect again, stay some time and try to connect again.

2- Steps for communication establishment between client and server

This chapter is not already complete. This is only a first version which will be filled out during development or sooner. It gives a first look of the protocol working. I think it should be enough to release a first reasonable version.

This part has been fully reexamined for removing the arranged in order point of view of packets transmitting. As you will see later, thanks to the study of numerous captures of packets with « tcpdump » on the 4661 and 4662 ports between an eDonkey2000 client and a p67 server, we've noticed that they both send packets in an untidy way. It could explain the need of using OPCODES which enables identification of packets and do without any pre-defined of for packets transmission.

2.1- Login Step

Client	Server
0x01 Hello <Client Info>	0x40 ID Change <Client ID>
0x15 Offer Files <File Info List>	0x38 Server Message [String]
0x14 Get Server List	0x32 Server List <Adress List>
	0x41 Server Info Data <Server Info>
	0x34 Server Status <Nusers><Nfiles>

2.2- Search File Step

Client	Server
0x16 Search File <Search Query>	0x33 Search File Results <File Info list><More>
0x21 More Results	

2.3- Search User Step

Client	Server
0x1a Search User <Search Query>	0x43 Search User results <Client Info List>

2.4- Get Sources Step

Client	Server
0x19 Get Sources <File Hash>	0x42 Found Sources <Files Hash><Adress List>

2.5- Callback Step

Client	Server
0x1c Client Callback Request <Client ID>	0x35 Server Callback Request <Client Adress>
	0x36 Callback Fail <Client ID>

eMule only uses a part of this step because in its OPCODES list, there are only the following ones : 0x35 et 0x1C which are the two first requests of this step. eMule asks for IP and port of corresponding to the client which ID was received in the <Client ID> packet. The server sends back all these informations and eMule is in charge of checking if the client is in its list or not. And if it finds this user in its list, it tries to connect to it, and if not, it creates the corresponding new entry in its users list and then tries to connect.

However I think that eMule tries to prevent the server from a doing certain part of the work because if 10 000 clients were doing a callback request for a Client ID and if the server had to manage all of them, you can easily imagine the traffic overload which will be generated.

I haven't finished to examine this step and particularly for eMule. It will be examined in more detail in the next documentation version.

3- Packets Contents

3.1- Data Units

BYTE = 1 octet
DWORD = 4 octets
WORD = 2 octets
FLOAT = float
HASH = 16 octets

[Client HASH] = HASH
[Server HASH] = HASH
[Packet Length] = DWORD
[File HASH] = HASH
[Client ID] = DWORD
[Port] = WORD
[IP] = DWORD
[Nusers] = DWORD
[Nfiles] = DWORD
[softlimit] = DWORD
[maxlimit] = DWORD
[maxUser] = DWORD

3.2- Different packets composition

All packets which have to be managed (because there are packets of which utility is not known and they don't have any of the following information) begin by the following information (after the TCP and IP headers) :

<Edonkey header>

<Packet Type> + [Packet Length]

<Packet Type>

0xe3 //for edonkey
or 0xc5 //for edonkey with eMule extensions
or 0xd4 //for eMule with compressed data

Then, there could be the following information :

<Client Info>

[Client Hash] + [Client ID] + [Port] + <Meta Tag List>

<File Info List>

DWORD + <File Info>*

<File Info>

[File Hash] + [Client ID] + [Port] + <Meta Tag List>

<Adress list>

BYTE + <Adress>*

<Adress>

[IP] + [PORT]

<Server Info>

[Server Hash] + [IP] + [Port] + <Meta Tag List>

<Server Status>

[Nusers] + [Nfiles]

<Meta Tag List>

DWORD + <Meta Tag>*

<Meta Tag>

0x02 + <Meta Tag Name> + [String]
or 0x03 + <Meta Tag Name> + DWORD
or 0x04 + <Meta Tag Name> + FLOAT

<Meta Tag Name>

[String]
or WORD + <Special Tag>*

<Special Tag>

Code	Contents	Type
0x01	[Name]	String
0x02	[Size]	DWORD
0x03	[Type] Audio, Video, ...	String
0x04	[Format] file extension	String
0x08	[Copied] contents already copied	DWORD
0x09	[GAP start]	DWORD
0x0a	[GAP end]	DWORD
0x0b	[Description]	String
0x0c	[Ping]	
0x0d	[Fail]	
0x0e	[Preference]	DWORD
0x0f	[Port]	WORD
0x10	[IP]	DWORD
0x11	[Version]	DWORD
0x12	[Tempfile] name of the temporary download file	
0x13	[Priority] 0 = low, 1 = normal et 2 = high	DWORD
0x14	[Status] 0 = looking, 1 = paused	DWORD
0x15	[Availablity]	DWORD
0x16	[QTime] this should mean Queue Time	
0x17	[Parts] called ULPRIORITY in eMule	
0x20	[Compression] eMule specific	
0x21	[Port UDP] UDP port for the client (eMule specific)	

This board is to be completed.

<Search Query>

<Operator> *+ <Parameter>*

<Parameter>

- 0x01 + [String] + <Meta Tag Name>
- or 0x02 + <Meta Tag>
- or 0x03 + DWORD <MinMax> + <Meta Tag Name>

<Operator>

- 0x00 (= AND)
- or 0x01 (= OR)
- or 0x02 (= AND NOT)

<MinMax>

- 0x01 (= min)
- or 0x02 (= max)

<More>

- 0x00 (No)
- or 0x01 (Yes)

4- UDP Communications

Here is a first outline of UDP communications between client and server. I think they are only used when the client is eMule.

<UDP Server status request>	0xE3+0x96+BYTE+BYTE+0xAA+0x5A
ou	0xE3+0x96+DWORD
<UDP Server status>	0xE3+0x97+BYTE+BYTE+0xAA+0x5A
	+ [Nusers]+[Nfiles]+[softlimit]+[hardlimit]+[maxUsers]
ou	0xE3+0x97+DWORD
	+ [Nusers]+[Nfiles]
<UDP Search file>	0xE3+0x98+<Search query>
<UDP Search file result>	0xE3+0x99+<File info>
<UDP Get sources>	0xE3+0x9A+<File hash>
<UDP Found sources>	0xE3+0x9B+<File hash> <Address list>
<UDP Callback request>	0xE3+0x9C+<Address> <Client ID>
<UDP Callback fail>	0xE3+0x9E+<Client ID>
<UDP ?>	0xE3+0xA0+?
<UDP Server list>	0xE3+0xA1+<Address list>
<UDP Get server info>	0xE3+0xA2
<UDP Server info>	0xE3+0xA3+<String : Server Name>
	<String : Server Description>
<UDP Get server list>	0xE3+0xA4

The ping value is calculated by the client when this one receives a 0x97 response next to a 0x96 request.

For the **<UDP Server status>** packet, BYTE+BYTE+0xAA+0x5A et DWORD are copies of **<UDP Server status request>** values.

5- Greetings

Many thanks to the pDonkey project of which you can find the sources on sourceforge.net. Its documentation has been very useful for doing this one as you will see.

Many thanks to the Switzern site of eDonkey 2000 : ed2k.ch, for their information about HighID and LowID.

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7- Changes

Version 3.1 :

- Chapter 2 : no more structures description
- Chapter 4 : changes in UDP communications

Version 3.0 :

- Chapter 3 : minor changes in the definition of the <Search Query> and the <Meta Tag>.
- Chapter 3 : The part about the HASH type at the end of the chapter has been removed.
- Chapter 3 : Special tags have been had in the board and some data units were removed.
- Chapter 4 : changes in the 0x97 UDP packet. Ping explication added.
- Chapter 1 : Explication of HighID/LowID has been re-written
- New chapter 6 : « Appendages » for the documentation's GNU license and chapter « Changes » has moved to chapter 7
- Changes in the first page.
- Changes in the page set.

Version 2.0 :

- Chapter 2 : addition of the TCPDUMP captures, and totally redone.
- Chapter 2 : addition of the C structures used by eFarm
- Chapter 3 : addition of the packets header.
- Chapter 4 was removed.
- Addition of the GPL License.

Version 1.1 :

- Addition of chapter 5 : special communications via UDP
- Addition of chapter 6 : I had forgotten it in first version, sorry.
- Addition of chapter 7 : Here we are, it's the second version.
- Addition of paragraphs 2.3 and 2.5 for « Search user » and « Callback » communication steps.
- Addition of some « Special Tag » in the board.

Version 1.0 : The first which comes along.